URBAN STRUCTURE

Lecture 12

- Urban Structure may be defined as a particular arrangement of adapted spaces, or lands in different uses, that might exist in urban area.

- The purpose of strategic land-use transport planning is to synthesize an urban structure which best houses the system of activities expected to develop in an urban area and which is capable of accommodating changes in these activity systems.

- A critical element of an urban-structure plan is the configuration of the transport network, since it ties together the other components of urban structure.

- The purpose of strategic land-use transport planning is to synthesize an urban structure which best houses the **system of activities**. The system of activities expected to develop in an urban area and which is capable of accommodating changes in these activity systems.

- This implies the land-use planners and transport system planners should be very clear about the system of activities in an urban area.

URBAN ACTIVITY SYSTEMS

In general, it has been observed that the bulk of activities occur in ordered hierarchies Certain activity centres such as elementary schools and small neighbourhood shopping centres exist at a scale that serves only local areas

Other types of urban activities exist in larger concentrations and may serve an entire urban region

The spread of the area of influence of different activities varies depending upon the type of activity

Examples of region-wide activities include major cultural facilities, post-secondary education facilities, specialised retail trade outlets and certain types of personal services

- The urban activity systems can be illustrated through idealization of the hierarchy of some of the activities located in urban areas
- Two activity systems are identified-

(i) household-workplace subsystem
(ii) household-service place subsystem

Urban Movement Hierarchies

Transport systems provide two fundamental functions –
movement and access to land-based activities

1. **Expressway** - Expressway provide for high volume and relatively fast movements. So, its capacity is more high volume lot of people can move faster simultaneously to and from major activity concentrations like the regional center

2. **Arterials - Arterials** provide for the movement of trips between Expressways and Collectors where ease of traffic movement is emphasized again and little or no direct access to land is provided
   Intersections between Arterials and Collectors are usually at grade and signalized

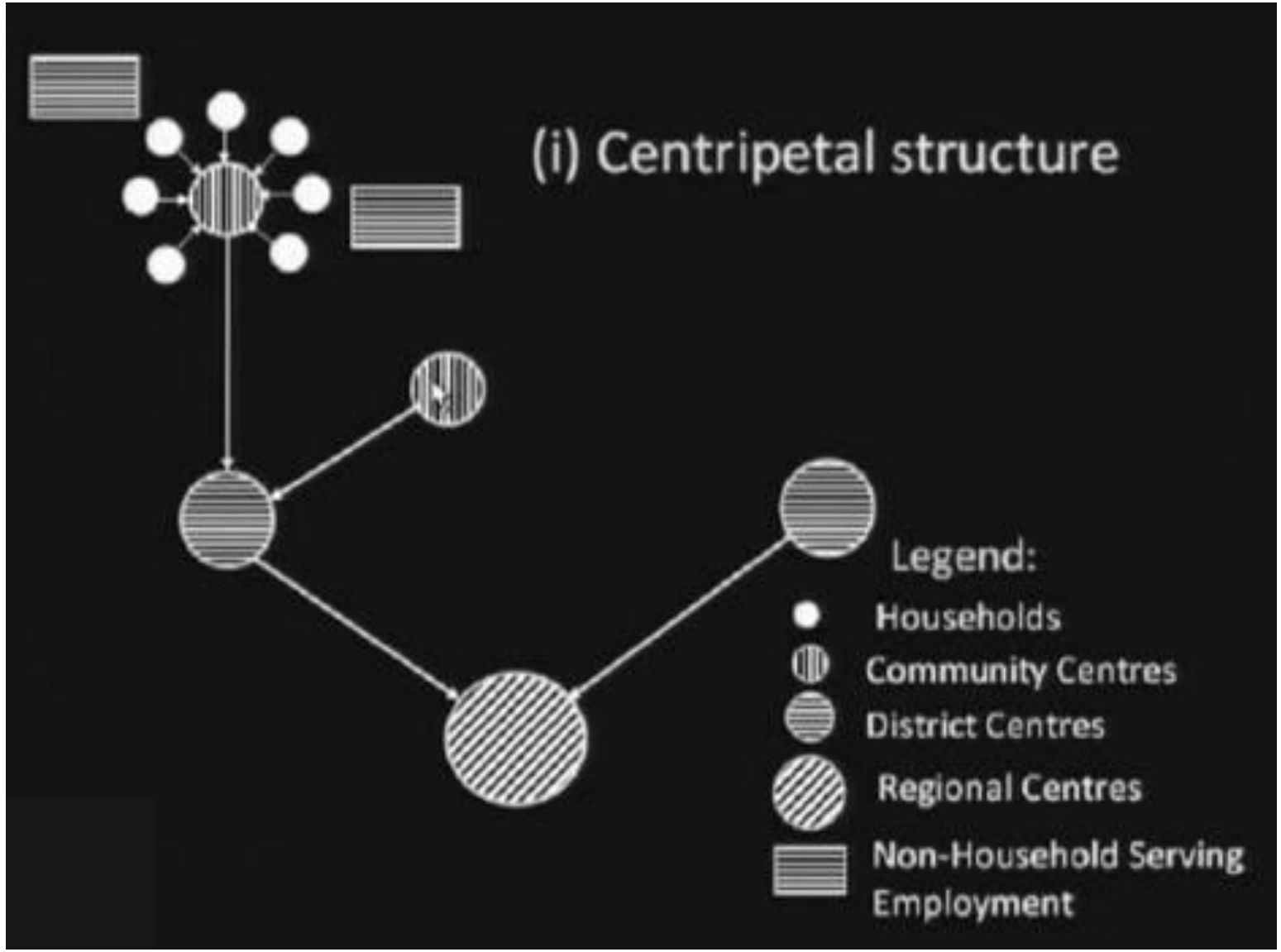
3. **Collector - Collector** streets provide for the movement of trips between Arterial and Locals and provide some direct access to land. Direct access to land from Collector is possible

4. Local streets - Local streets provide for the distribution of traffic within activity areas, that is the primary function distribution of traffic within activity

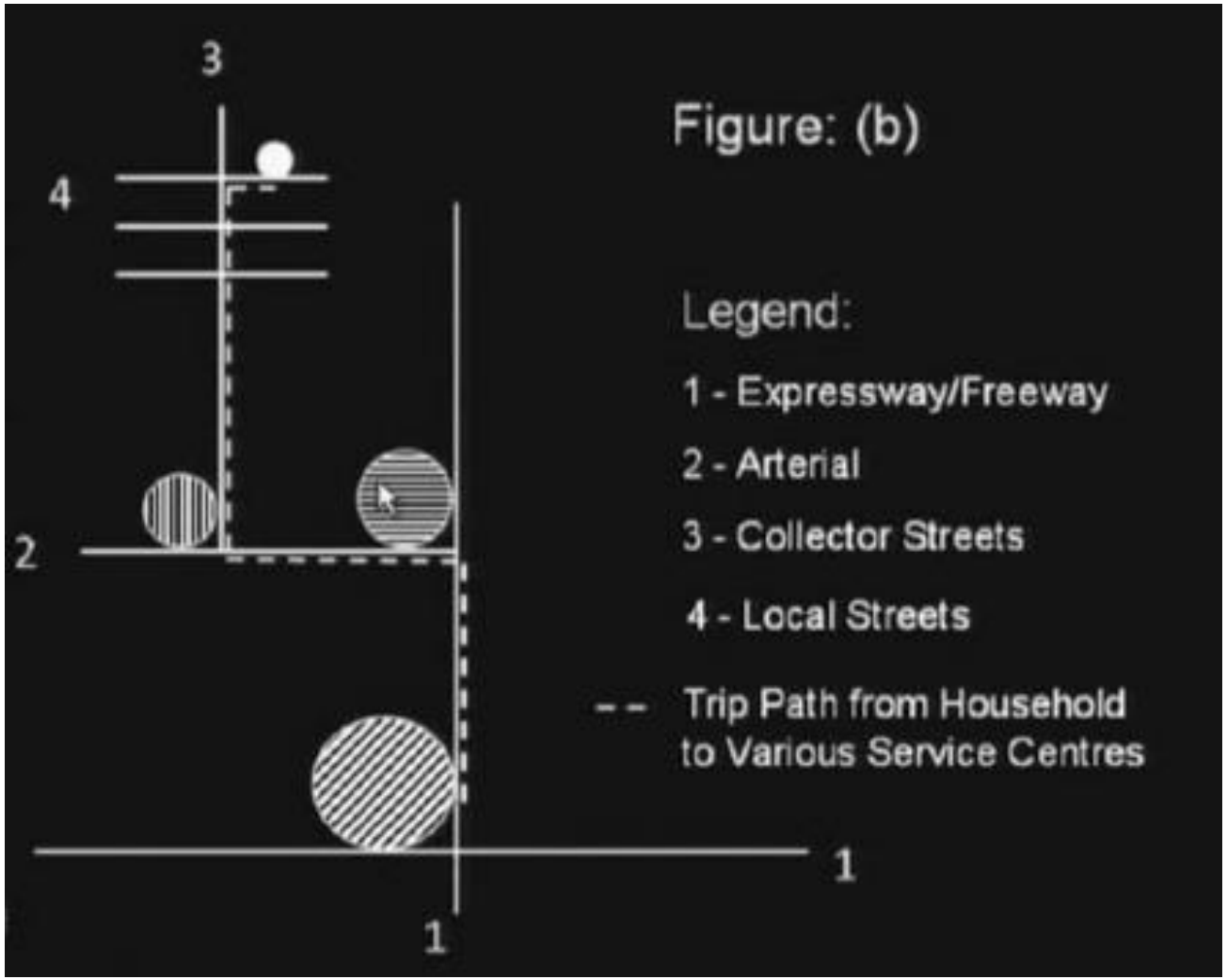
TYPES OF URBAN STRUCTURE

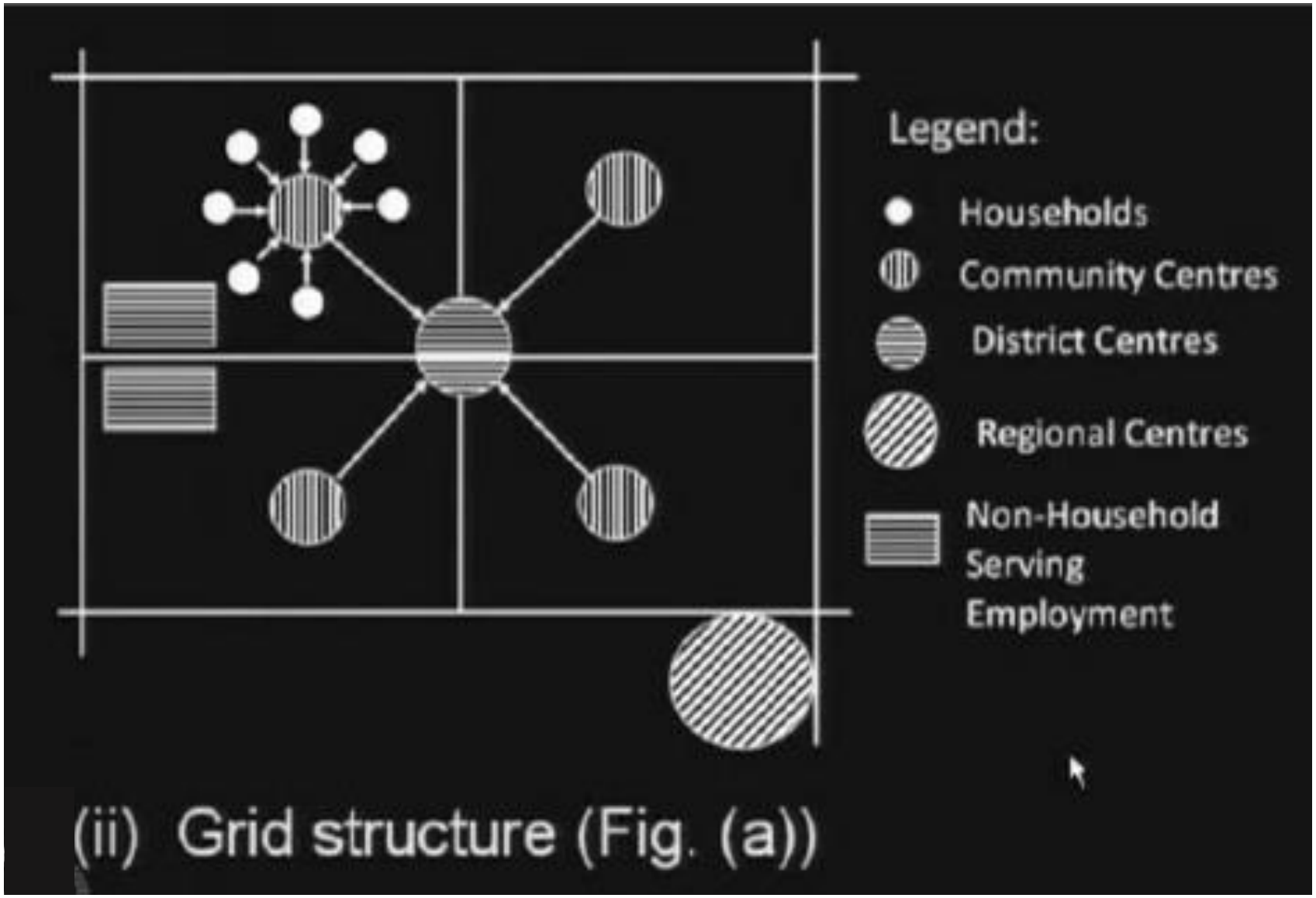
Basically, there are three types of urban structure:

(i) Centripetal structure
(ii) Grid Structure
(iii) Linear Structure


(i) Centripetal structure
Legend:
Households
Community Centres
District Centres
Regional Centres
Non-Household Serving Employment

Fitness of transport system hierarchy with the activity centre hierarchy




Legend:
Households
Community Centres
District Centres
Regional Centres
Non-Household Serving Employment
(ii) Grid structure (Fig. (a))

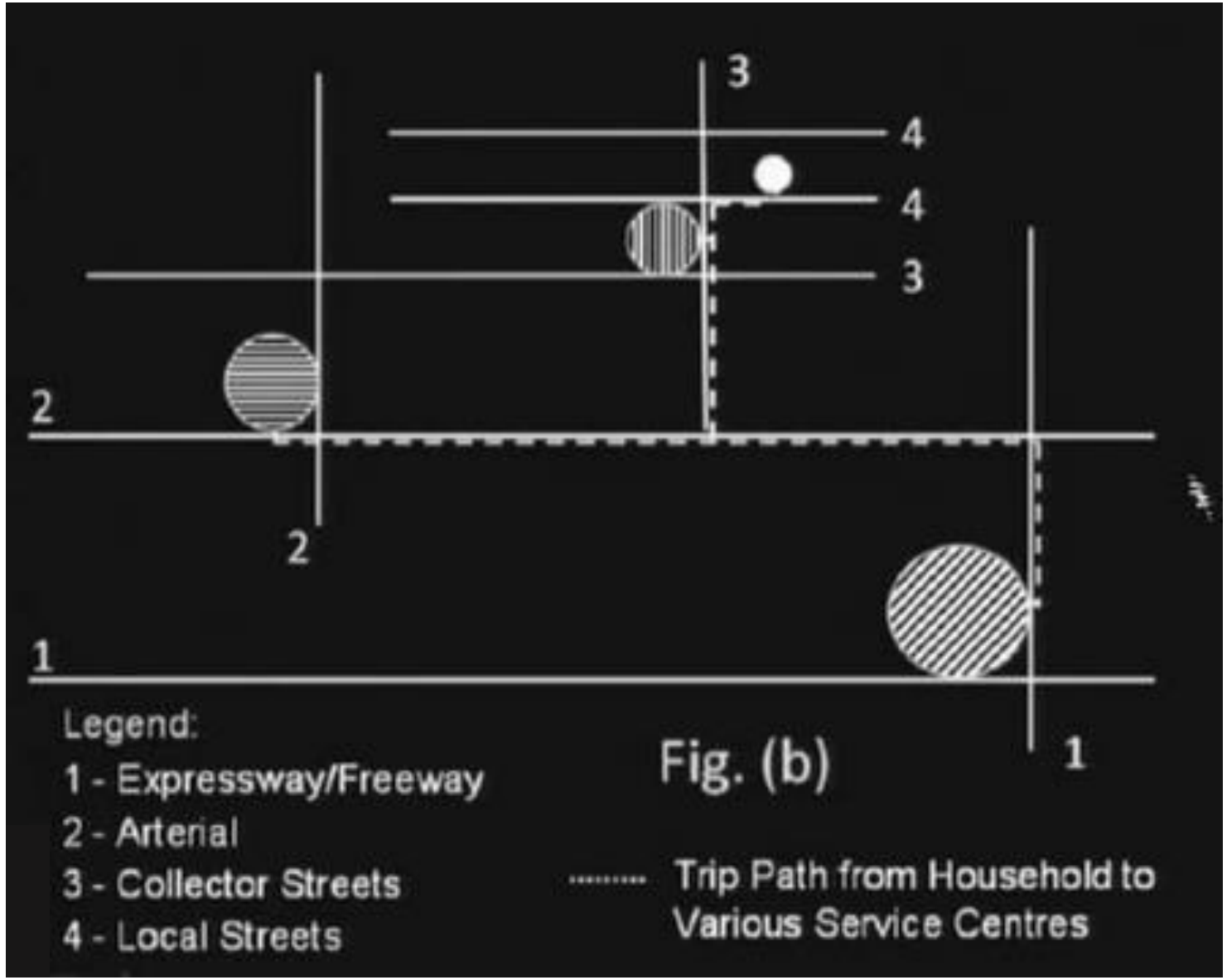
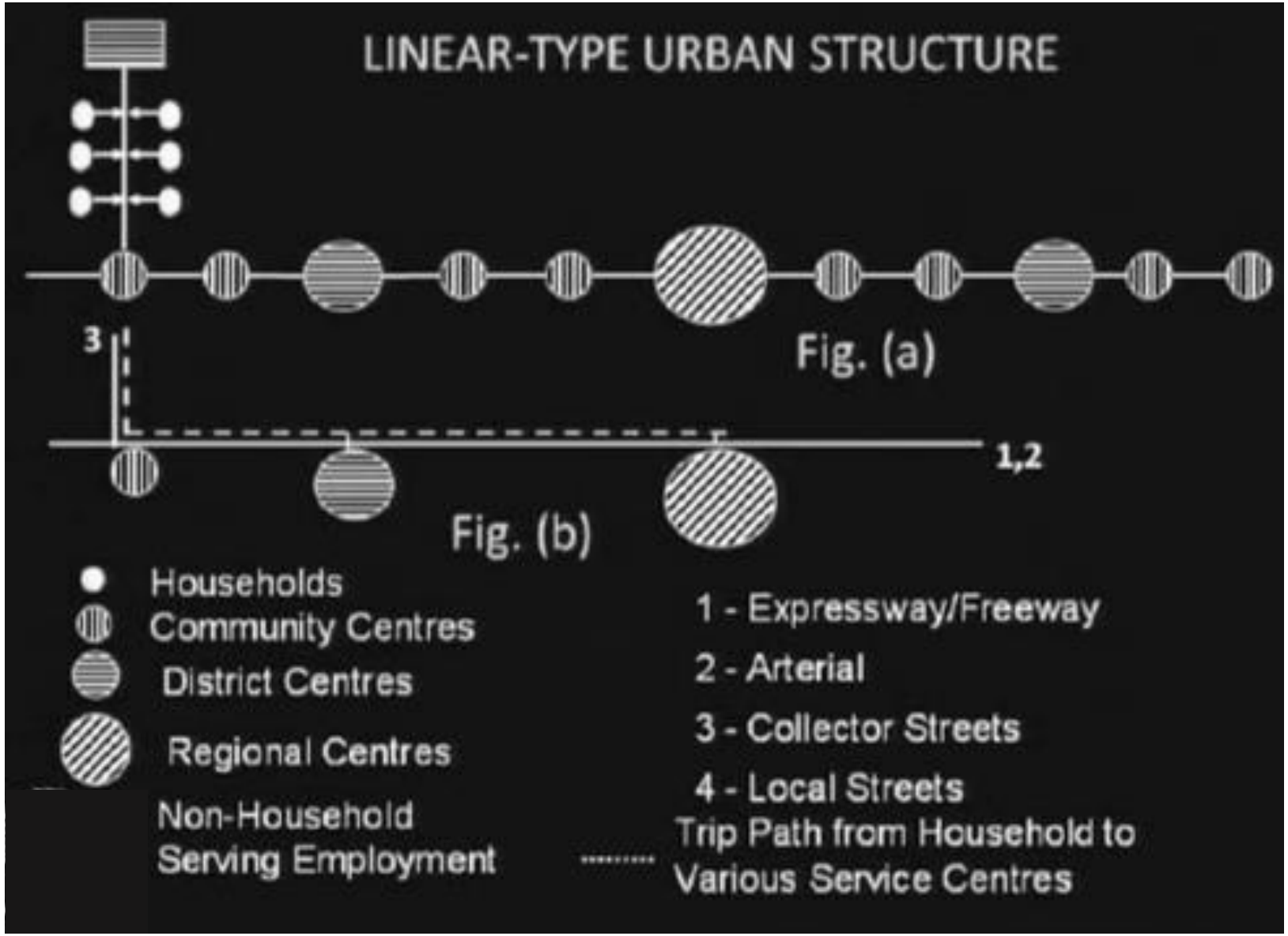

3
4
4
3
2
2
1
1
Legend:
1 - Expressway/Freeway
2 - Arterial
3 - Collector Streets
4 - Local Streets
Trip Path from Household to Various Service Centres

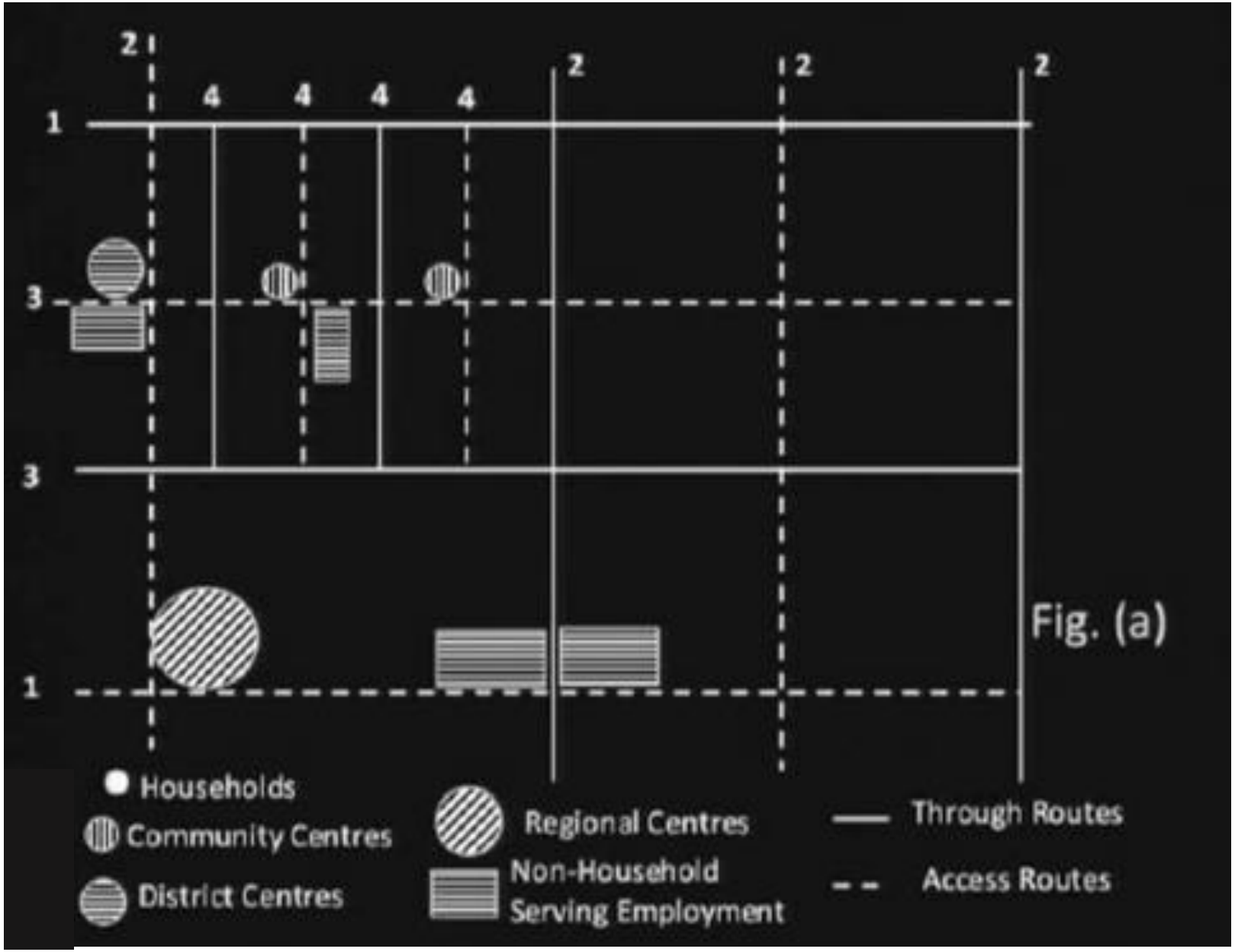
Fig. (b)


LINEAR-TYPE URBAN STRUCTURE
Fig. (a)
3
1,2
Fig. (b)
Households
Community Centres
District Centres
Regional Centres
Non-Household
Serving Employment
1 - Expressway/Freeway
2 - Arterial
3 - Collector Streets
4 - Local Streets
Trip Path from Household to
Various Service Centres

Directional Grid Urban Structure

- This structure is called the directional grid urban structure and represents a synthesis of some of the principles contained within the three concepts discussed earlier

- The basis of this concept is a more specialised transport route hierarchy



SUSTAINABILITY

- A strategic planning instrument of the local level to address specific local transport and mobility problems
  - Air quality and CO_2- reduction
  - Congestion
  - Traffic safety
  - Urban sprawl
  - Accessibility
- Improvement of overall quality of urban life and competitiveness
- Balanced mix of modes with focus on environmentally sound and low carbon modes
- Integrated development of transport infrastructure (road, rail, water) towards multi- and intermodality

Traditional Transport Planning		Sustainable Urban Mobility Planning
Focus on traffic	→	Focus on people
Primary objectives: Traffic flow capacity and speed	→	Primary objectives: Accessibility and quality of life, as well as sustainability, economic viability, social equity, health and environmental quality
Modal-focussed	→	Balanced development of all relevant transport modes and shift towards cleaner and more sustainable transport modes
Infrastructure focus	→	Integrated set of actions to achieve cost-effective solutions
Sectorial planning document	→	Sectorial planning document that is consistent and complementary to related policy areas (such as land use and spatial planning; social services; health; enforcement and policing; etc.)
Short- and medium-term delivery plan	→	Short- and medium-term delivery plan embedded in a long-term vision and strategy
Related to an administrative area	→	Related to a functioning area based on travel-to-work patterns
Domain of traffic engineers	→	Interdisciplinary planning teams
Planning by experts	→	Planning with the involvement of stakeholders using a transparent and participatory approach
Limited impact assessment	→	Regular monitoring and evaluation of impacts to inform a structured learning and improvement process

Goals and objectives

A Sustainable Urban Mobility Plan has as its central goal improving accessibility of urban areas and providing high-quality and sustainable mobility and transport to, through and within the urban area. It regards the needs of the 'functioning city' and its hinterland rather than a municipal administrative region.

In pursuit of this goal, a Sustainable Urban Mobility Plan seeks to contribute to development of an urban transport system which:

(a.) Is accessible and meets the basic mobility needs of all users;

(b.) Balances and responds to the diverse demands for mobility and transport services by citizens, businesses and industry;

(c.) Guides a balanced development and better integration of the different transport modes;

(d.) Meets the requirements of sustainability, balancing the need for economic viability, social equity, health and environmental quality;

(e.) Optimises efficiency and cost effectiveness;

(f.) Makes better use of urban space and of existing transport infrastructure and services;

(g.) Enhances the attractiveness of the urban environment, quality of life, and public health;

(h.) Improves traffic safety and security;

(i.) Reduces air and noise pollution, greenhouse gas emissions, and energy consumption; and